Race Condition

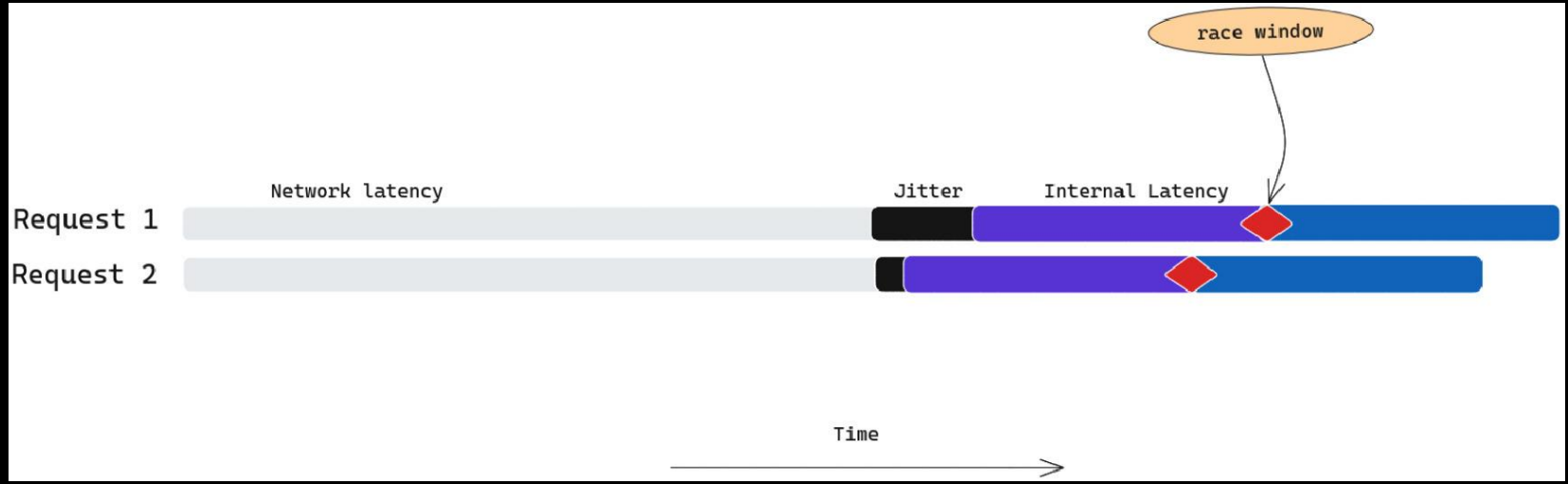
Race Condition

- Why?
  - *Non-Atomic Operations*
- What is Race Condition?
- How to detect?
- How to fix Race Condition?

Useful Commands:

- To check race condition window
  - *while true; do go run program.go; done*
- To detect race conditions
  - *Go run -race program.go*

Race Condition Window

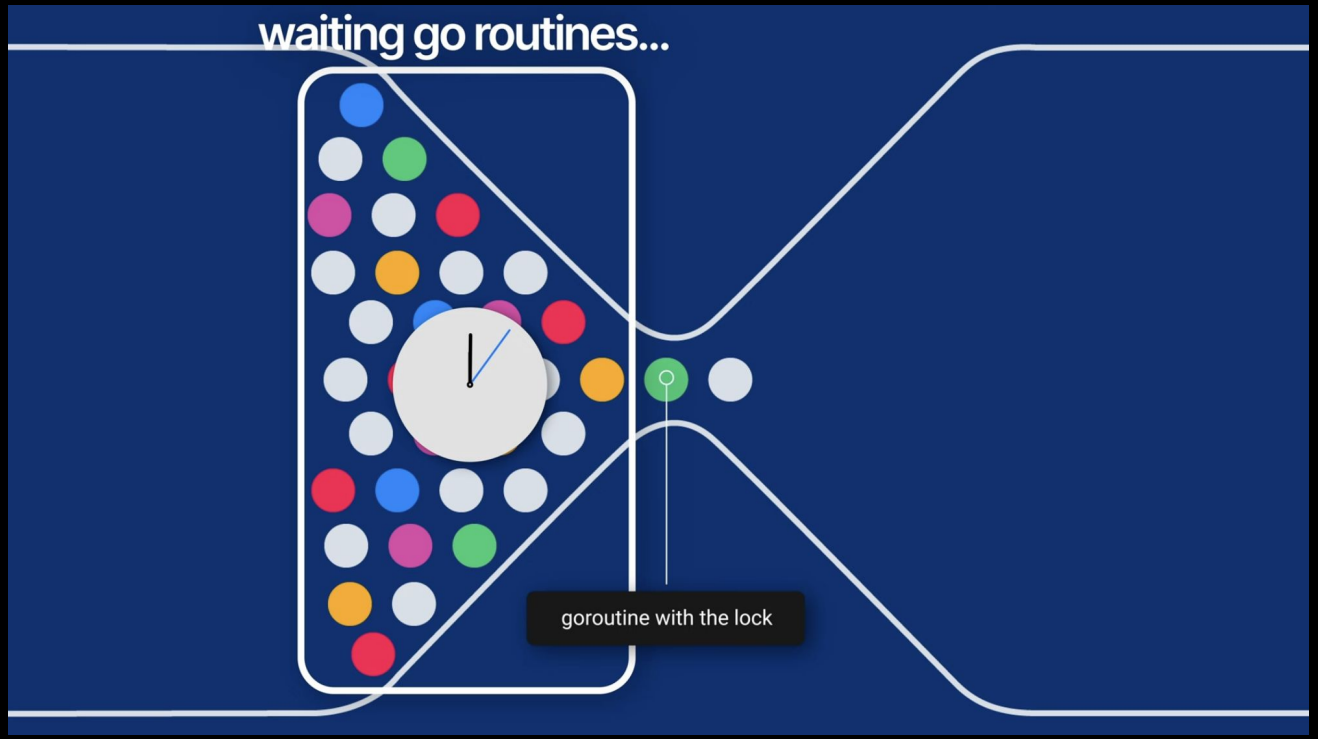


Race Condition Window

Mutex

- A mutex **(mutual exclusion lock)** ensures that only one goroutine accesses **critical section** at a time
- Used to protect **shared mutable state**
- Critical Section
  - ***Any code that reads + writes*** *shared data must be protected*

```go
var mu sync.Mutex

mu.Lock()
// critical section
mu.Unlock()
```

Mutex